Homogeneous equations

We've looked at how to use the substitution $v = y^{1-n}$ to change a Bernoulli equation into a linear equation, but this isn't the only equation type we can solve with a change of variable.

Form of a homogeneous differential equation

We'll also use a change of variable to solve homogeneous equations, which are differential equations in the form

$$y' = F\left(\frac{y}{x}\right)$$

In other words, this type of differential equation can be solved for y' (or dy/dx), and is given in terms of some function defined in y/x. For instance,

$$y' = 1 + \frac{y}{x}$$

is an example of a simple homogeneous equation. It's solved for y', and written in terms of y/x.

Keep in mind that homogeneous equations won't always be written in a way that they are obviously homogeneous. We'll often have to do some manipulation with the equation in order to see that it's homogeneous.

Confirming homogeneity

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If it's not immediately obvious that the equation is homogeneous, we do have one check we can do to help us determine the homogeneity of the equation. First, we need to make sure that the equation is solved for dy/dx, or y'.

Then we can look to see whether every term on the opposite side of the equation has the same degree. For instance, looking at the right side of

$$y' = \frac{-4x^2 - y^2}{xy}$$

we see three terms, $-4x^2$, $-y^2$, and xy. Both $-4x^2$ and $-y^2$ are second-degree terms, since they both have an exponent of 2. The xy term is also a second-degree term, since x and y both have degree 1, which adds up to $1 + 1 = 2$.

As long as these terms all have the same degree, whether they're all second-degree, all third-degree, etc., then we know the differential equation is homogeneous.

Alternately, we can also confirm homogeneity if $f(\lambda x, \lambda y) = f(x, y)$. Using

$$y' = \frac{-4x^2 - y^2}{xy}$$

the equation is homogeneous because

$$f(\lambda x, \lambda y) = \frac{-4(\lambda x)^2 - (\lambda y)^2}{(\lambda x)(\lambda y)}$$

$$= \frac{-4\lambda^2 x^2 - \lambda^2 y^2}{\lambda^2 xy}$$

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$$= \frac{\lambda^2(-4x^2 - y^2)}{\lambda^2 xy}$$

$$= \frac{-4x^2 - y^2}{xy} = f(x, y)$$

Because we were able to confirm that $f(\lambda x, \lambda y) = f(x, y)$, we can say that the differential equation is homogeneous.

Solving homogeneous equations

Once we have the homogeneous equation written in standard form, then our goal will be to use a change of variable to rewrite it as a separable differential equation.

Then we'll find the solution to the separable equation, and finally we'll back substitute in order to find the solution to the homogeneous equation.

In summary, we'll follow these steps:

1. If the homogeneous equation isn't already given in standard form, rewrite it in standard form.
2. Substitute $v = y/x$ and $y' = v + xv'$.
3. Solve the resulting separable equation.
4. Back-substitute for $v = y/x$.
5. If possible, solve explicitly for y to find the solution to the homogeneous differential equation.

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Let's work through an example so that we can practice solving homogeneous equations with a change of variable.

Example

Find the solution to the differential equation.

$$xyy' + 2x^2 - 3y^2 = 0$$

To put the equation in standard form, we'll divide through by x^2.

$$\frac{xy}{x^2}y' + \frac{2x^2}{x^2} - 3\frac{y^2}{x^2} = 0$$

$$\frac{y}{x}y' + 2 - 3\left(\frac{y}{x}\right)^2 = 0$$

$$\frac{y}{x}y' = 3\left(\frac{y}{x}\right)^2 - 2$$

and then solve for y'.

$$y' = 3\left(\frac{y}{x}\right)^2\left(\frac{x}{y}\right) - 2\left(\frac{x}{y}\right)$$

$$y' = \frac{3}{\frac{y}{x}}\left(\frac{y}{x}\right)^2 - \frac{2}{\frac{y}{x}}$$

$$y' = 3\left(\frac{y}{x}\right) - \frac{2}{\frac{y}{x}}$$

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Now that the equation is in standard form, we'll substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = 3v - \frac{2}{v}$$

$$v^2 + xvv' = 3v^2 - 2$$

$$xvv' = 2v^2 - 2$$

Now we should have a separable differential equation, so we'll separate variables,

$$xv\frac{dv}{dx} = 2v^2 - 2$$

$$xv\ dv = (2v^2 - 2)\ dx$$

$$\frac{v}{2v^2 - 2}\ dv = \frac{1}{x}\ dx$$

and then integrate both sides, using the substitution $u = 2v^2 - 2$, $du = 4v\ dv$, and $dv = (1/4v)\ du$.

$$\int \frac{v}{2v^2 - 2}\ dv = \int \frac{1}{x}\ dx$$

$$\int \frac{v}{u}\left(\frac{1}{4v}\right)\ du = \int \frac{1}{x}\ dx$$

$$\frac{1}{4}\int \frac{1}{u}\ du = \int \frac{1}{x}\ dx$$

$$\frac{1}{4}\ln|u| = \ln|x| + C$$

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Back substitute with $u = 2v^2 - 2$.

$$\frac{1}{4}\ln|2v^2-2| = \ln|x| + C$$

$$\ln|2v^2-2| = 4\ln|x| + 4C$$

$$e^{\ln|2v^2-2|} = e^{4\ln|x|+4C}$$

$$|2v^2-2| = e^{4\ln|x|}e^{4C}$$

$$2v^2-2 = Ce^{4\ln|x|}$$

$$2v^2-2 = Ce^{\ln|x|^4}$$

$$2v^2-2 = C|x|^4$$

$$2v^2 = C|x|^4 + 2$$

$$v^2 = C|x|^4 + 1$$

$$v^2 = Cx^4 + 1$$

Then we back substitute with $v = y/x$.

$$\left(\frac{y}{x}\right)^2 = Cx^4 + 1$$

$$\frac{y^2}{x^2} = Cx^4 + 1$$

$$y^2 = Cx^6 + x^2$$

$$y = \pm\sqrt{Cx^6 + x^2}$$

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Let's work through another example to get some more practice with solving homogeneous equations.

Example

Use a substitution to find a solution to the homogeneous equation.

$$y' = \frac{-4x^2 - y^2}{xy}$$

We'll multiply through the numerator and denominator on the right side by $1/(x^2)$.

$$y' = \frac{-4x^2 - y^2}{xy} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$y' = \frac{-4x^2\left(\frac{1}{x^2}\right) - y^2\left(\frac{1}{x^2}\right)}{xy\left(\frac{1}{x^2}\right)}$$

$$y' = \frac{-4 - \frac{y^2}{x^2}}{\frac{y}{x}}$$

$$y' = \frac{-4 - \left(\frac{y}{x}\right)^2}{\frac{y}{x}}$$

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Substitute $v = y/x$ and $y' = v + xv'$.

$$v + xv' = \frac{-4 - v^2}{v}$$

Now we should have a separable differential equation, so we'll separate variables,

$$x\frac{dv}{dx} = \frac{-4 - v^2}{v} - v$$

$$x\frac{dv}{dx} = \frac{-4 - 2v^2}{v}$$

$$xv\ dv = (-4 - 2v^2)\ dx$$

$$\frac{v}{2v^2 + 4}\ dv = -\frac{1}{x}\ dx$$

and then integrate both sides, using a simple substitution for the integral on the left, with $u = v^2 + 2$ and $dv = du/2v$.

$$\frac{1}{2}\int \frac{v}{v^2 + 2}\ dv = -\int \frac{1}{x}\ dx$$

$$\frac{1}{2}\int \frac{v}{u} \cdot \frac{du}{2v} = -\int \frac{1}{x}\ dx$$

$$\frac{1}{4}\ln u = -\ln x + C$$

$$\frac{1}{4}\ln(v^2 + 2) = -\ln x + C$$

Back-substitute for v.

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$$\frac{1}{4}\ln\left(\left(\frac{y}{x}\right)^2+2\right)=-\ln x+C$$

$$\ln\left(\left(\frac{y}{x}\right)^2+2\right)=-4\ln x+C$$

$$\ln\left(\frac{y^2}{x^2}+2\right)=-4\ln x+C$$

$$e^{\ln\left(\frac{y^2}{x^2}+2\right)}=e^{-4\ln x+C}$$

$$\frac{y^2}{x^2}+2=Ce^{-4\ln x}$$

$$\frac{y^2}{x^2}+2=Ce^{\ln(x^{-4})}$$

$$\frac{y^2}{x^2}+2=Cx^{-4}$$

$$y^2+2x^2=Cx^{-2}$$

$$y^2=\frac{C}{x^2}-2x^2$$

$$y^2=\frac{C-2x^4}{x^2}$$

We'll allow this to be the implicitly-defined solution to the homogeneous equation.

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